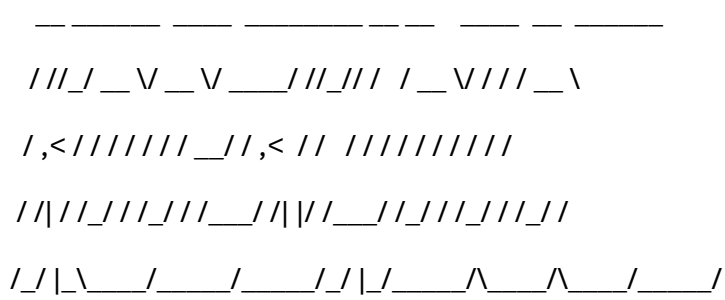
Welcome to the KodeKloud Hands-On lab



~ ➜ ls

Dockerfile-mysql Dockerfile-ubuntu app

Dockerfile-python Dockerfile-wordpress

~ ➜ cat Dockerfile-mysql

FROM debian:stretch-slim

add our user and group first to make sure their IDs get assigned consistently, regardless of whatever dependencies get added

RUN groupadd -r mysql && useradd -r -g mysql mysql

RUN apt-get update && apt-get install -y --no-install-recommends gnupg dirmngr && rm -rf /var/lib/apt/lists/*

add gosu for easy step-down from root

ENV GOSU_VERSION 1.7

RUN set -x \

```
    && apt-get update && apt-get install -y --no-install-recommends ca-certificates wget && rm -rf /var/lib/apt/lists/* \
    && wget -O /usr/local/bin/gosu "https://github.com/tianon/gosu/releases/download/$GOSU_VERSION/gosu-$(dpkg --print-architecture)" \
    && wget -O /usr/local/bin/gosu.asc "https://github.com/tianon/gosu/releases/download/$GOSU_VERSION/gosu-$(dpkg --print-architecture).asc" \
    && export GNUPGHOME="$(mktemp -d)" \
    && gpg --batch --keyserver ha.pool.sks-keyservers.net --recv-keys B42F6819007F00F88E364FD4036A9C25BF357DD4 \
    && gpg --batch --verify /usr/local/bin/gosu.asc /usr/local/bin/gosu \
    && gpgconf --kill all \
    && rm -rf "$GNUPGHOME" /usr/local/bin/gosu.asc \
    && chmod +x /usr/local/bin/gosu \
    && gosu nobody true \
    && apt-get purge -y --auto-remove ca-certificates wget

RUN mkdir /docker-entrypoint-initdb.d

RUN apt-get update && apt-get install -y --no-install-recommends \
# for MYSQL_RANDOM_ROOT_PASSWORD
        pwgen \
# for mysql_ssl_rsa_setup
        openssl \
# FATAL ERROR: please install the following Perl modules before executing /usr/local/mysql/scripts/mysql_install_db:
# File::Basename
```

```
# File::Copy

# Sys::Hostname

# Data::Dumper

        perl \

    && rm -rf /var/lib/apt/lists/*


RUN set -ex; \

# gpg: key 5072E1F5: public key "MySQL Release Engineering <mysql-build@oss.oracle.com>" imported

    key='A4A9406876FCBD3C456770C88C718D3B5072E1F5'; \

    export GNUPGHOME="$(mktemp -d)"; \

    gpg --batch --keyserver ha.pool.sks-keyservers.net --recv-keys "$key"; \

    gpg --batch --export "$key" > /etc/apt/trusted.gpg.d/mysql.gpg; \

    gpgconf --kill all; \

    rm -rf "$GNUPGHOME"; \

    apt-key list > /dev/null


ENV MYSQL_MAJOR 8.0

ENV MYSQL_VERSION 8.0.17-1debian9


RUN echo "deb http://repo.mysql.com/apt/debian/ stretch mysql-${MYSQL_MAJOR}" > /etc/apt/sources.list.d/mysql.list


# the "/var/lib/mysql" stuff here is because the mysql-server postinst doesn't have an explicit way to disable the mysql_install_db codepath besides having a database already "configured" (ie, stuff in /var/lib/mysql/mysql)

# also, we set debconf keys to make APT a little quieter
```

```
RUN { \
        echo mysql-community-server mysql-community-server/data-dir select ''; \
        echo mysql-community-server mysql-community-server/root-pass password ''; \
        echo mysql-community-server mysql-community-server/re-root-pass password ''; \
        echo mysql-community-server mysql-community-server/remove-test-db select false; \
    } | debconf-set-selections \
    && apt-get update && apt-get install -y mysql-community-client="${MYSQL_VERSION}" mysql-community-server-core="${MYSQL_VERSION}" && rm -rf /var/lib/apt/lists/* \
    && rm -rf /var/lib/mysql && mkdir -p /var/lib/mysql /var/run/mysqld \
    && chown -R mysql:mysql /var/lib/mysql /var/run/mysqld \
# ensure that /var/run/mysqld (used for socket and lock files) is writable regardless of the UID our mysqld instance ends up having at runtime
    && chmod 777 /var/run/mysqld

VOLUME /var/lib/mysql

# Config files

COPY config/ /etc/mysql/

COPY docker-entrypoint.sh /usr/local/bin/

RUN ln -s usr/local/bin/docker-entrypoint.sh /entrypoint.sh # backwards compat

ENTRYPOINT ["docker-entrypoint.sh"]

EXPOSE 3306 33060

CMD ["mysqld"]

~ ➜ ls

Dockerfile-mysql    Dockerfile-ubuntu    app
```

Dockerfile-python Dockerfile-wordpress

~ ➜ cat Dockerfile-wordpress

```
FROM php:7.1-apache

# install the PHP extensions we need (https://make.wordpress.org/hosting/handbook/handbook/server-environment/#php-extensions)
RUN set -ex; \
    \
    savedAptMark="$(apt-mark showmanual)"; \
    \
    apt-get update; \
    apt-get install -y --no-install-recommends \
        libjpeg-dev \
        libmagickwand-dev \
        libpng-dev \
    ; \
    \
    docker-php-ext-configure gd --with-png-dir=/usr --with-jpeg-dir=/usr; \
    docker-php-ext-install -j "$(nproc)" \
        bcmath \
        exif \
        gd \
        mysqli \
        opcache \
```

```
        zip \
    ; \
    pecl install imagick-3.4.4; \
    docker-php-ext-enable imagick; \
    \
# reset apt-mark's "manual" list so that "purge --auto-remove" will remove all build dependencies
    apt-mark auto '.*' > /dev/null; \
    apt-mark manual $savedAptMark; \
    ldd "$(php -r 'echo ini_get("extension_dir");')"/*.so \
        | awk '/=>/ { print $3 }' \
        | sort -u \
        | xargs -r dpkg-query -S \
        | cut -d: -f1 \
        | sort -u \
        | xargs -rt apt-mark manual; \
    \
    apt-get purge -y --auto-remove -o APT::AutoRemove::RecommendsImportant=false; \
    rm -rf /var/lib/apt/lists/*

# set recommended PHP.ini settings
# see https://secure.php.net/manual/en/opcache.installation.php
RUN { \
        echo 'opcache.memory_consumption=128'; \
        echo 'opcache.interned_strings_buffer=8'; \
        echo 'opcache.max_accelerated_files=4000'; \
```

```
        echo 'opcache.revalidate_freq=2'; \
        echo 'opcache.fast_shutdown=1'; \
    } > /usr/local/etc/php/conf.d/opcache-recommended.ini
# https://wordpress.org/support/article/editing-wp-config-php/#configure-error-logging
RUN { \
# https://www.php.net/manual/en/errorfunc.constants.php
# https://github.com/docker-library/wordpress/issues/420#issuecomment-517839670
        echo 'error_reporting = E_ERROR | E_WARNING | E_PARSE | E_CORE_ERROR | E_CORE_WARNING | E_COMPILE_ERROR | E_COMPILE_WARNING | E_RECOVERABLE_ERROR'; \
        echo 'display_errors = Off'; \
        echo 'display_startup_errors = Off'; \
        echo 'log_errors = On'; \
        echo 'error_log = /dev/stderr'; \
        echo 'log_errors_max_len = 1024'; \
        echo 'ignore_repeated_errors = On'; \
        echo 'ignore_repeated_source = Off'; \
        echo 'html_errors = Off'; \
    } > /usr/local/etc/php/conf.d/error-logging.ini

RUN a2enmod rewrite expires

VOLUME /var/www/html

ENV WORDPRESS_VERSION 5.2.2
ENV WORDPRESS_SHA1 3605bcbe9ea48d714efa59b0eb2d251657e7d5b0
```

```
RUN set -ex; \
    curl -o wordpress.tar.gz -fSL "https://wordpress.org/wordpress-${WORDPRESS_VERSION}.tar.gz"; \
    echo "$WORDPRESS_SHA1 *wordpress.tar.gz" | sha1sum -c -; \
# upstream tarballs include ./wordpress/ so this gives us /usr/src/wordpress
    tar -xzf wordpress.tar.gz -C /usr/src/; \
    rm wordpress.tar.gz; \
    chown -R www-data:www-data /usr/src/wordpress

COPY docker-entrypoint.sh /usr/local/bin/

ENTRYPOINT ["docker-entrypoint.sh"]
CMD ["apache2-foreground"]
```

```
~ ➜ ls
Dockerfile-mysql     Dockerfile-ubuntu    app
Dockerfile-python    Dockerfile-wordpress

~ ➜ cat Dockerfile-ubuntu
#
# Ubuntu Dockerfile
#
# https://github.com/dockerfile/ubuntu
#

# Pull base image.
```

```
FROM ubuntu:14.04

# Install.
RUN \
  sed -i 's/# \(.*multiverse$\)/\1/g' /etc/apt/sources.list && \
  apt-get update && \
  apt-get -y upgrade && \
  apt-get install -y build-essential && \
  apt-get install -y software-properties-common && \
  apt-get install -y byobu curl git htop man unzip vim wget && \
  rm -rf /var/lib/apt/lists/*

# Add files.
ADD root/.bashrc /root/.bashrc
ADD root/.gitconfig /root/.gitconfig
ADD root/.scripts /root/.scripts

# Set environment variables.
ENV HOME /root

# Define working directory.
WORKDIR /root

# Define default command.
CMD ["bash"]
```

~ ➜ docker run -d ubuntu sleep 1000

db3679183d1c5d3c0290d8cfb8cdbe4a368901d4953f79499a754586292107bf